

Customer Churn Prediction Using Machine Learning

Final AI/ML Project — Week 4

1. Introduction

Customer churn refers to a customer stopping their use of a company's service. Predicting churn can help businesses identify customers who may leave and take retention actions.

2. Problem Statement

Build a binary classification system that predicts whether a customer will churn based on customer demographic, subscription, payment, usage, support and satisfaction information.

3. Objectives

Clean the dataset, perform EDA, preprocess features, train two classification models, compare their performance, save the final model, and provide a Streamlit prediction interface.

4. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib, Jupyter Notebook and Streamlit.

5. Dataset Description

The provided dataset contains 15,100 rows and 17 columns before duplicate removal. The target variable is churn. customer_id is excluded from model training because it is an identifier.

6. Data Preprocessing

Missing numerical values are handled with median imputation. Missing categorical values are handled with most-frequent imputation. Numerical features are standardized and categorical features are one-hot encoded.

7. Exploratory Data Analysis

The notebook includes target distribution, numerical distributions, churn comparisons by categorical variables and correlation analysis. The target contains 12,559 non-churn and 2,541 churn customers.

8. Model Development

Two classification algorithms were trained: Logistic Regression and Random Forest Classifier. An 80:20 stratified train/test split was used.

9. Results

Logistic Regression achieved Accuracy 0.9303, Precision 0.7565, Recall 0.8631, F1-Score 0.8063 and ROC-AUC 0.9454. Random Forest achieved Accuracy 0.9340, Precision 0.8097, Recall 0.7937, F1-Score 0.8016 and ROC-AUC 0.9379.

10. Model Selection

Logistic Regression was selected as the final model based primarily on F1-Score because the dataset is imbalanced and churn detection requires a balance between precision and recall.

11. Challenges Faced

Handling missing values, categorical encoding, class imbalance, multiple evaluation metrics and maintaining consistent preprocessing during deployment were the main challenges.

12. Future Improvements

Future work can include hyperparameter tuning, cross-validation, threshold optimization, SHAP explainability, more customer history, real-time data integration and cloud deployment.

13. Conclusion

An end-to-end customer churn prediction system was successfully developed. The project covers data cleaning, EDA, preprocessing, model training, evaluation, comparison, model saving and deployment preparation.

Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9303	0.7565	0.8631	0.8063	0.9454
Random Forest	0.9340	0.8097	0.7937	0.8016	0.9379